

Threats to Identification

ECON 5200 Final Project: Causal Effect of ACA Medicaid Expansion on County-Level Uninsured Rates

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1. Most Serious Threat: Violation of Parallel Trends

The core identifying assumption of difference-in-differences is that, absent the ACA expansion, expansion and non-expansion counties would have followed parallel trajectories in uninsurance.

This assumption is fundamentally untestable in the post-treatment period. While our event study shows pre-2014 differences statistically indistinguishable from zero, and a placebo difference-in-differences using a fake 2012 treatment date returns a null coefficient ($\beta = 0.25$, $p = 0.21$), these tests speak only to pre-period parallelism. Expansion states differ systematically from non-expansion states along multiple dimensions: they have higher median incomes, more Democratic political control, more generous pre-ACA safety nets, more mature insurance exchanges, and stronger healthcare provider networks. Any of these factors could plausibly generate differential post-2014 coverage trajectories independent of Medicaid expansion itself.

Direction of bias: away from zero (overstating the causal effect). Expansion states' broader infrastructure advantages plausibly drove uninsurance down faster regardless of the Medicaid policy. For example, expansion states were more likely to operate their own state-based insurance exchanges with active outreach, which would lower uninsured rates through marketplace coverage rather than Medicaid. Our DiD estimate attributes this exchange-related decline to expansion.

What would address this threat: An ideal design would exploit *within-state* variation around the 138% Federal Poverty Level eligibility threshold using individual-level American Community Survey data — a regression discontinuity that does not require comparing different states. Alternatively, the synthetic control method (Abadie, Diamond, & Hainmueller 2010) could construct a counterfactual trajectory for each expansion state from a weighted combination of non-expansion states, weighting on pre-treatment outcomes rather than assuming a simple group average suffices.

2. Spillover Effects (SUTVA Violation)

The Stable Unit Treatment Value Assumption requires that one unit's treatment does not affect another unit's outcome. Two violations are plausible. First, individuals near state borders may migrate to expansion states to gain coverage, simultaneously reducing the treatment group's uninsured rate (through the migrants who enroll) and reducing the control group's uninsured rate (by exporting uninsured residents). Second, ACA-related media attention may have driven

"woodwork effects" — previously eligible individuals enrolling in *traditional* Medicaid even in non-expansion states — through a channel correlated with but not caused by the expansion provision itself.

Direction of bias: toward zero (understating the true expansion effect). If border migration moves uninsured individuals from control to treated states, control-state uninsured rates fall artificially, shrinking the DiD estimate. The woodwork effect operates similarly: control states see coverage gains attributable to ACA salience, which we incorrectly net out as part of the secular trend.

What would address it: A border-pair design (Dube, Lester, & Reich 2010) comparing adjacent counties on opposite sides of state expansion borders would more cleanly isolate the policy effect from migration and salience-driven spillovers. Our exclusion of staggered adopters partially mitigates the most severe Goodman-Bacon (2021) concerns but does not address SUTVA per se.

3. Confounding ACA Provisions

Our "treatment" variable captures more than the Medicaid expansion provision in isolation. The ACA bundled multiple major provisions effective in 2014: the individual mandate, federally- and state-run insurance exchanges, premium subsidies, pre-existing condition protections, and expanded dependent coverage. Expansion states tended to engage in more aggressive overall ACA implementation — running their own exchanges, conducting outreach campaigns, and supplementing federal subsidies — which we cannot separately identify from the Medicaid mechanism.

Why this matters: Our estimate is best interpreted as the combined effect of Medicaid expansion *plus correlated state-level ACA implementation*, not the marginal effect of Medicaid alone. For state policymakers deciding whether to adopt the full ACA expansion package today, this bundled estimate remains the relevant policy parameter; for narrower questions about Medicaid mechanism design, it is not.

4. Specification Choice and Small-Cluster Inference

Our unweighted TWFE estimate (−2.06 pp, 95% CI: [−3.42, −0.69]) and population-weighted estimate (−1.41 pp, 95% CI: [−3.81, 0.99]) differ materially. The unweighted specification gives equal weight to a 500-person rural county and a five-million-person urban county; the weighted specification reports the average effect on the average person. Both are legitimate but answer different policy questions, and the weighted CI crosses zero.

Separately, our standard errors cluster at the state level — appropriate because treatment is state-assigned (Abadie, Athey, Imbens, & Wooldridge 2017) — but with only 35 state clusters, we sit at the lower edge of where cluster-robust inference is reliable. Cameron and Miller (2015) suggest 40+ clusters as a rule of thumb. A wild cluster bootstrap (Cameron, Gelbach, & Miller 2008) would provide more reliable inference under this constraint and is a clear avenue for robustness extension.

5. External Validity to Remaining Non-Expansion States

The 10 states still not expanded by 2023 — Alabama, Florida, Georgia, Kansas, Mississippi, South Carolina, Tennessee, Texas, Wisconsin, and Wyoming — are systematically different from the 2014 expanders that identify our estimate. They are poorer, more rural, more Republican-governed, and have weaker pre-existing healthcare infrastructure. Whether the policy effect estimated on 2014 expanders generalizes to these states depends on whether the response function is homogeneous across these dimensions, which seems unlikely.

Direction of bias for policy projection: Likely *toward zero* if applied uncritically. Lower take-up rates in states with weaker outreach infrastructure would attenuate the realized effect. Our Streamlit dashboard makes this explicit through a $0.5\text{--}1.5\times$ generalizability multiplier, allowing decision-makers to scenario-test under conservative external-validity assumptions.

6. What I Cannot Rule Out

Despite five pre-treatment years, parallel-trend evidence in the event study, a null placebo DiD, county and year fixed effects, and state-clustered standard errors, I cannot fully rule out non-parallel secular trends correlated with the political choice to expand. The estimate should be interpreted as **the combined effect of Medicaid expansion and associated state-level policy uptake on uninsurance, among states that chose to expand in 2014** — not as a context-free, generalizable Medicaid effect. The honest scope of inference is narrower than the headline estimate suggests, and the consulting recommendation should be framed accordingly.

7. Personal Reflection

Working through this project deepened my understanding of what difference-in-differences identification actually delivers. The most important lesson for me was recognizing that the so-called parallel-trends “tests” — the event study and the placebo DiD — only verify parallelism in the *past*. They cannot guarantee that trends would have continued to be parallel in the post-period absent the policy. No matter how many pre-treatment years I have, I cannot rule out that some independent factor began accelerating uninsurance reduction in expansion states starting in 2014.

This made me realize that intellectual honesty in causal inference is not measured by statistical p-values; it is measured by clear-eyed awareness of where the identifying assumption could fail.

Of all the threats discussed above, the one I personally find most concerning is SUTVA violation through cross-border population movement. Migration is hard to fully observe in aggregate panel data, and it biases the estimate toward zero. If I had more time, I would implement a border-pair design (Dube, Lester, & Reich 2010) as a robustness check, comparing adjacent counties on opposite sides of expansion-state borders. This project taught me that causal inference is not about producing a single “gold standard” number; it is the cumulative result of a series of design choices, each of which involves judgment about the direction and magnitude of potential biases.

References

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